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Subject: Machine Learning

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EXPERIMENT 8

AIM: To implement Decision Tree.

Code:

```
import pandas as pd

from sklearn.tree import DecisionTreeClassifier

from sklearn import tree

import matplotlib.pyplot as plt


# 1. Prepare the dataset
# Manual dataset (Outlook example)
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain', 'Overcast', 'Sunny'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Mild', 'Cool'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong', 'Weak'],
    'PlayTennis': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No']
}

df = pd.DataFrame(data)

# Convert categorical data to numeric using one-hot encoding
# We drop the 'PlayTennis' column as it's our target variable
df_encoded = pd.get_dummies(df.drop('PlayTennis', axis=1))

# Target variable (what we want to predict)
y = df['PlayTennis']

print("Original DataFrame:")
```



```
display(df)
```

```
print("Encoded Features (df_encoded):")
```

```
display(df_encoded.head())
```

```
print("Target Variable (y):")
```

```
display(y.head())
```

```
# 2. Create and Train a Decision Tree Classifier
```

```
# We'll use the Gini impurity as the criterion for splitting nodes
```

```
model = DecisionTreeClassifier(criterion='gini', random_state=42) # random_state for reproducibility
```

```
# Train the model using the encoded features and the target variable
```

```
model.fit(df_encoded, y)
```

```
print("Decision Tree Model trained successfully using Gini impurity.")
```

```
# 3. Make a Prediction
```

```
# Let's predict for the first sample in our dataset
```

```
sample_to_predict = df_encoded.iloc[0:1]
```

```
prediction = model.predict(sample_to_predict)
```

```
print(f"Sample to predict:\n{sample_to_predict}")
```

```
print(f"Prediction for this sample: {prediction[0]}")
```

```
# You can also get prediction probabilities
```

```
probabilities = model.predict_proba(sample_to_predict)
```

```
print(f"Prediction probabilities: {probabilities}")
```



4. Visualize the Decision Tree

```
plt.figure(figsize=(15, 10)) # Adjust figure size for better readability

tree.plot_tree(
    model,
    feature_names=df_encoded.columns, # Names of the features
    class_names=['No', 'Yes'],      # Names of the target classes
    filled=True,                    # Fill nodes with colors to indicate class
    rounded=True,                   # Round node boxes
    fontsize=10                     # Adjust font size
)

plt.title("Decision Tree trained on 'PlayTennis' data (Gini Impurity)", fontsize=16)

plt.show()
```

Output:

Original DataFrame:

	Outlook	Temperature	Humidity	Wind	PlayTennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Mild	Normal	Strong	Yes
7	Sunny	Cool	High	Weak	No

Encoded Features (df_encoded):

	Outlook_Overcast	Outlook_Rain	Outlook_Sunny	Temperature_Cool	Temperature_Hot	Temperature_Mild	Humidity_High	Humidity_Normal	Wind_Strong	Wind_Weak
0	False	False	True	False	True	False	True	False	False	True
1	False	False	True	False	True	False	True	False	True	False
2	True	False	False	False	True	False	True	False	False	True
3	False	True	False	False	False	True	True	False	False	True
4	False	True	False	True	False	False	False	True	False	True

Target Variable (y):

	PlayTennis
0	No
1	No
2	Yes
3	Yes
4	Yes

dtype: object



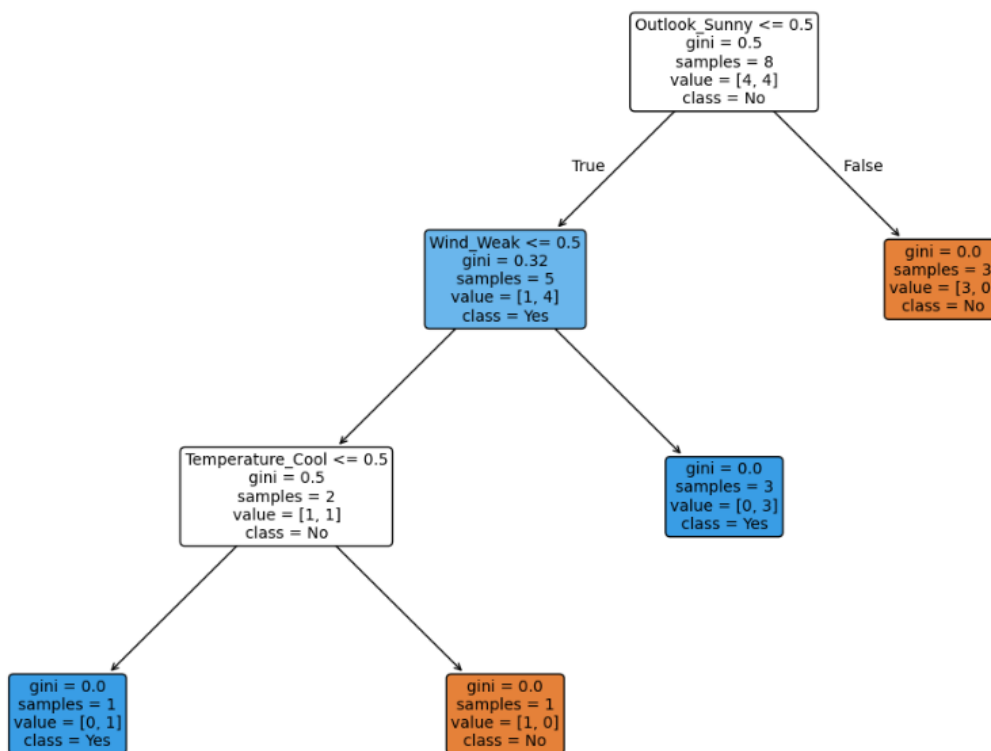
... Decision Tree Model trained successfully using Gini impurity.

```
Sample to predict:
  Outlook_Overcast Outlook_Rain Outlook_Sunny Temperature_Cool \
0           False      False           True           False

  Temperature_Hot Temperature_Mild Humidity_High Humidity_Normal \
0           True           False           True           False

  Wind_Strong Wind_Weak
0          False      True
Prediction for this sample: No
Prediction probabilities: [[1. 0.]]
```

Decision Tree trained on 'PlayTennis' data (Gini Impurity)



Conclusion:

Thus, I have successfully implemented Decision Tree Classifier.